

THE PRESIGNAL FIELD NODE v2.0

A Triggered Multi-Spectrum Acquisition System for Geological Plasma Discharge Research

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ABSTRACT

We describe the Presignal Field Node v2.0 — a triggered multi-spectrum acquisition system extending the Presignal EM Sentinel electromagnetic monitoring platform with an integrated camera activation layer. The Field Node v2.0 operates across four sequential layers: a continuously active electromagnetic sentinel layer monitoring ten sensor channels derived from the Geological Pathway Diversity Model (GPDM) framework; a trigger scoring engine that calculates a weighted composite score from simultaneous sensor inputs and activates camera acquisition when threshold is crossed; a directed capture layer comprising visible, thermal infrared, near-ultraviolet, and spectral imaging channels mounted on a motorized equatorial mount; and an AI classification pipeline that performs automated prosaic elimination and GPDM pathway classification on captured imagery correlated with simultaneous electromagnetic data. Camera channel selection is informed by the finding of Galan Santos and Koren (2024) that the majority of discharge events are invisible at visible wavelengths, making thermal infrared a primary detection channel rather than a supplementary one. The complete system deploys for approximately \$1,788 USD and is released under CC BY 4.0.

Keywords: geological plasma discharge, triggered acquisition, APP detection, earth lights, GPDM, electroball, ball lightning, Hessdalen, multi-spectrum imaging, ULF triangulation, AI classification

1. INTRODUCTION

The systematic scientific investigation of anomalous aerial plasma phenomena (APP) faces a fundamental instrumentation challenge. Optical camera systems can document the visual behavior of discharge phenomena but cannot detect events below the visible threshold and consume power continuously. Electromagnetic monitoring systems detect geophysical signatures of discharge precursors and events continuously at low power cost, but cannot provide visual documentation of the resulting plasma phenomena.

Galan Santos and Koren (2024) identify a critical detection constraint with direct design implications: the majority of electroballs are transparent or invisible at visible wavelengths and detectable only via infrared cameras or polarized filters. This finding means that a camera system monitoring only the visible spectrum will miss the majority of discharge events. Thermal IR capture is therefore not supplementary but a primary detection requirement for sub-visual events. Existing field monitoring deployments at documented APP sites, including Project Hessdalen, have operated camera and electromagnetic systems independently, resulting in temporal gaps between electromagnetic anomaly detection and optical documentation.

The Field Node v2.0 resolves this by integrating the EM Sentinel electromagnetic monitoring package (Carter, 2026a) with a triggered camera acquisition system that activates only when the electromagnetic trigger score crosses a threshold indicating a probable discharge event. This ensures continuous power-efficient monitoring while guaranteeing that any geophysically significant event, including sub-visual events detectable only in

thermal IR, produces simultaneous multi-spectrum optical documentation.

2. SYSTEM ARCHITECTURE

The Field Node v2.0 operates across four integrated layers. Scientific value derives from simultaneous multi-channel timestamp correlation across all layers during a confirmed event.

Layer	Name	Function	Power	Data Produced
1	EM Sentinel	Continuous 10-channel GPDM monitoring. Rolling trigger score at 10Hz.	Always on ~3W	Continuous timestamped 10-channel EM record. The baseline that makes every trigger interpretable.
2	Trigger Engine	When score reaches 60 points, fires GPIO relay chain to wake cameras and mount. Latency target: under 500ms.	Fires on threshold	Trigger log with timestamp, score, and channel contributions. Documents EM environment at acquisition start.
3	Directed Capture	Motor mount slews to calculated source direction. All four camera channels capture simultaneously.	Active during events ~25W	Simultaneous IR + visible + UV + spectral image sequence, GPS-timestamped and correlated with EM trigger record.
4	AI Classification	Post-capture prosaic elimination, EM correlation, multi-spectrum fusion, GPDM pathway classification, upload.	Post-event ~3W	Classified event package with confidence score and full correlated EM + optical dataset ready for human review.

Core design principle: electromagnetic sensors watch continuously at negligible power cost. Cameras activate only when electromagnetic evidence indicates something worth filming has already been detected. Every camera activation is accompanied by a pre-existing EM record of what triggered it.

3. TRIGGER SCORE LOGIC

No single sensor fires the acquisition. The system calculates a rolling 30-second trigger score from weighted sensor inputs. When the score crosses 60 points, acquisition fires. Each sensor weight reflects both its reliability as a discharge indicator and the scientific value of a positive reading.

Sensor Event	Score	Discharge Indicator Rationale	Scientific Value if This Fires
ULF unipolar pulse >100 pT	40	Unipolar signature distinguishes geological from lightning (bipolar). Primary p-hole current indicator per Freund (2011).	Highest-value single-channel trigger. Confirmed unipolar pulse correlated with camera capture is the primary electromagnetic evidence of geological discharge.
EFM sudden collapse from elevation	60	Direct real-time discharge indicator. EFM collapse releases accumulated atmospheric charge. Fires acquisition alone.	Real-time discharge documentation. Combined with thermal IR, simultaneously captures EM and thermal signature of the discharge moment.
EFM sustained elevation >500 V/m	35	Pre-discharge charge accumulation. Sustained elevation indicates a building event not yet discharged.	Pre-event capture. Camera activation during elevation without collapse may document the pre-discharge plasma formation phase.
Ground stakes >20mV sustained	35	Direct p-hole surface current measurement. The specific charge carrier mechanism proposed by Freund (2011).	Pathway 1 direct evidence. Stake activation correlated with camera capture provides surface-to-atmosphere current documentation.
Ion count >1000 ions/cm ³	20	Atmospheric ionization field building. State Variable 3 elevation indicating charge accumulation.	Pre-event atmospheric state. Ion density at trigger time enables comparison with plasma emission spectrum from AS7341.
Geiger elevation >50% above baseline	20	Radon or ionization field elevation. Pathway 5 indicator or solar particle precipitation.	Pathway 5 state documentation. Elevated CPM correlated with visual capture allows comparison of radiation and optical emission environments.
Ozone >80 ppb	15	Active corona discharge chemical byproduct. May precede visible discharge.	Chemical pre-event indicator. Ozone elevation at trigger may document pre-discharge corona before visual threshold.

Infrasound tonal 4-8 Hz	15	Geological stress acoustic emission. Direct State Variable 1 measurement.	Mechanical state documentation. Infrasound correlated with visual capture provides acoustic evidence of geological stress driving the event.
Accelerometer micro-seismic burst	10	Microseismic activity correlated with ULF pulse events.	Mechanical-to-electromagnetic coupling evidence for p-hole pathway.
TRIGGER THRESHOLD	60	Requires EFM collapse alone, or ULF + ground stakes, or multiple medium-weight channels. Single-sensor noise cannot fire.	Every acquisition has a defined multi-channel EM justification on record.

4. DIRECTED CAPTURE — CAMERA ARRAY SPECIFICATION

Four imaging channels activate simultaneously on trigger. Channel selection is built around the Galan Santos and Koren (2024) finding that most electroballs are invisible at visible wavelengths. The thermal IR channel is therefore designated PRIMARY — it is the only channel that documents sub-visual discharge events. Each channel captures what the others cannot.

Channel	Sensor	Wavelength	Cost	Detection Role	What This Channel Uniquely Captures
Thermal IR PRIMARY	FLIR Lepton 3.5 + PureThermal 3	8,000-14,000 nm	\$299	Primary: sub-visual discharge events; ground thermal anomalies	The only channel that detects events invisible to the naked eye and visible camera. Per Galan Santos & Koren (2024), transparent electroballs are detectable only via IR. Captures plasma thermal footprint even when no optical emission occurs. Essential — not optional.
Visible / NIR	Sionyx Aurora Pro	380-1000 nm	\$599 or existing	Visual events; plasma color and morphology; low-light events	Gen3-equivalent sensitivity documents APP behavior and color at visual threshold and below in NIR. Captures plasma morphology and trajectory directly comparable to historical Hessdalen documentation and eyewitness accounts.
Near UV	RPi HQ Camera + Baader U-filter	320-380 nm	\$185	Nitrogen and oxygen plasma emission lines; events with UV output but minimal visible light	N2 and O2 UV emission lines are bright even when total visible output is low. Documents events missed by both IR and visible channels. Third independent spectral domain for cross-validation.
Spectral AS7341	AMS AS7341 11-channel	415-680 nm + NIR	Incl. in Sentinel	Plasma mineral composition fingerprint at moment of any trigger event	Wavelength-resolved emission at trigger moment regardless of camera pointing accuracy. Identifies plasma mineral composition — geological pathway indicator from optical data alone. Produces scientific data even when directed cameras miss the event due to pointing error.
Wide Field Context	RPi Camera Module 3 Wide	Autofocus wide	\$38	Event spatial context; motor mount pointing verification	Continuous context ensures documentation even if motor mount slew is imprecise. Documents spatial relationship between event and terrain for source localization cross-reference with stake directional data.

4.1 Data Yield per Event Scenario

The following table defines what data the Field Node v2.0 produces under different event scenarios, ensuring that every trigger activation produces a defined and useful dataset.

Event Scenario	EM Channels Active	Camera Output	Scientific Data in Hand	Evidential Weight
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Sub-visual EM only (no thermal or visible output)	ULF + EFM collapse or ground stakes	IR: possible faint anomaly. Visible: nothing. UV: nothing. AS7341: ambient baseline.	Multi-channel EM discharge signature GPS-timestamped. Thermal IR provides negative or weak result. AS7341 provides ambient emission baseline at moment of event.	Class B-C event. EM signature alone publishable as case report. Strengthens with repetition at same site.
Sub-visual thermal event (IR anomaly, no visible output)	ULF + EFM + ground stakes elevated	IR: clear thermal anomaly. Visible: nothing. UV: possibly faint. AS7341: possible weak emission.	Multi-channel EM + thermal IR documentation of discharge event invisible to the naked eye. The most scientifically significant scenario — direct evidence of sub-visual discharge corroborated by independent EM channels across three domains.	Class A event. EM + IR + AS7341 across three independent domains. Publishable with baseline context.
Full visual event (visible + EM + IR + UV)	All primary detection channels elevated	IR: clear anomaly. Visible: plasma morphology documented. UV: emission lines. AS7341: mineral fingerprint.	Four-domain dataset: EM multi-channel + thermal IR + visible morphology + UV emission + mineral fingerprint. GPS timestamp links to USGS seismic and NASA DONKI catalogs for external forcing context.	Class A maximum evidential weight. Tier-one geophysics journal target with 6+ month baseline.
Background deployment (no trigger events)	All channels logging continuously	No camera activation. Wide-field context continuous if powered.	12-month EM baseline, groundwater EC precursor record, seasonal ion and ozone variation, GPS ionospheric proxy, Forbush Decrease catalog correlation.	Essential scientific infrastructure. A 12-month clean baseline from an unmonitored candidate site is a publishable contribution independent of event capture.

5. SOURCE DIRECTION CALCULATION

Two ground stake pairs are deployed perpendicular to each other — one North-South, one East-West. The differential voltage provides a vector indicating probable source direction of the p-hole surface current, with accuracy of approximately 30-45 degrees of arc with two pairs, improving to 15-20 degrees with a third pair at 45 degrees. This directional estimate determines the motor mount slew target at trigger. The wide-field context camera provides pointing verification independent of the stake calculation — if directed cameras miss the event due to pointing error, the wide-field frame documents the event location relative to the field of view, enabling retrospective calibration for subsequent events.

In a dual-node configuration with two Field Node v2.0 systems deployed 200-400 meters apart, the microsecond difference in ULF pulse arrival time between magnetometers enables significantly more precise triangulation of source direction, approximately 5-15 degrees of arc. This is the primary scientific rationale for the Hessdalen dual-node configuration specified in Section 7.

6. COMPLETE BILL OF MATERIALS

Component	Cost USD	Category	Notes
Complete EM Sentinel package (Carter 2026a)	\$516	EM monitoring	All ten electromagnetic channels. Layer 1 continuous monitoring and trigger score.
Second Raspberry Pi Zero 2W (acquisition controller)	\$15	Core	Dedicated Pi for camera and mount control. Keeps EM logging uninterrupted during acquisition.
4-channel relay module + MOSFET switching array	\$14	Trigger hardware	GPIO-controlled power switching. Sub-500ms activation latency target.
Marine-grade fuse block + USB-C PD trigger board	\$27	Trigger hardware	Protects camera power circuits from surge at activation.

Sky-Watcher Star Adventurer GTi motorized mount	\$399	Motor mount	Azimuth and altitude via WiFi API. Target position within 3 seconds of trigger. Direction from stake differential.
Mount heater wrap 12V PTC	\$22	Thermal	Prevents lubricant failure below -10 degrees C.
FLIR Lepton 3.5 + PureThermal 3 board	\$299	IR channel PRIMARY	Primary detection for sub-visual events. 8-14 um thermal IR. Required per Galan Santos and Koren (2024) detection protocol.
Raspberry Pi HQ Camera + Baader U-filter	\$185	UV channel	320-380 nm near-UV. Nitrogen and oxygen plasma emission lines. Third independent spectral domain.
Sionyx Aurora Pro night vision camera	\$599 or existing	Visible/NIR channel	Gen3-equivalent sensitivity. Primary visual documentation. Already owned by many investigators.
Raspberry Pi Camera Module 3 Wide	\$38	Wide field context	Continuous context and motor mount pointing verification.
Camera mounting plate Arca-Swiss standard	\$18	Hardware	Universal camera attachment.
Extended LiFePO4 40Ah battery + BMS	\$95	Power	Larger capacity for camera power draw. MANDATORY LiFePO4 chemistry.
100W solar panel (upgraded from Sentinel 20W)	\$85	Power	Camera array requires more power budget. Undersized panel causes acquisition failures in cloudy periods.
Additional IP67 enclosure for camera electronics	\$35	Enclosure	Separate weatherproof housing for camera control electronics.
Cabling, connectors, conduit, hardware	\$25	Misc	22 AWG wire throughout. Stainless steel hardware only.
TOTAL	~\$1,788 USD	-	Without Sionyx Aurora Pro: ~\$1,189.

7. HESSDALEN DUAL-NODE DEPLOYMENT CONFIGURATION

The Hessdalen Valley in Norway represents the optimal first deployment target for the Field Node v2.0. It is the most documented persistent APP site in the world, has existing monitoring infrastructure from Project Hessdalen, and its geology qualifies under GPDM Pathway 4 (natural battery mechanism) and Pathway 6 (seismoelectric/hydroelectric coupling in a confined aquifer valley).

A dual-node configuration with two Field Node v2.0 systems deployed 200-400 meters apart enables geometric source localization through ULF pulse timing triangulation between two magnetometers, achieving approximately 5-15 degrees of arc accuracy versus 30-45 degrees from a single-node stake geometry. All pairwise comparisons can be cross-referenced against the existing Project Hessdalen geological survey data.

The dual-node Hessdalen experiment would produce, to our knowledge, the first triangulated source position for an APP event globally. A confirmed triangulated source position correlated with geological pathway data and GPDM state variable conditions at time of event constitutes the evidence required to move the framework from correlation to causation — the necessary step for publication in a tier-one geophysics journal. The simultaneous four-domain optical dataset from both nodes would provide the most comprehensive single-event APP documentation yet produced.

8. AI CLASSIFICATION PIPELINE

Post-capture AI processing performs five sequential stages before human review is required. The pipeline maximizes scientific yield by ensuring the human review package contains a fully correlated, fully contextualized dataset rather than raw imagery requiring manual multi-channel correlation.

Stage	Input	Process	Output for Human Review
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1 — Prosaic Elimination	Wide-field frame + GPS timestamp + weather API + TLE satellite database + ADS-B aircraft feed	Match event against aircraft strobe patterns, TLE-predicted satellite positions, star catalog, lightning bipolar EM signature, vehicle horizontal source.	Elimination log with confidence score per prosaic category. Unclaimed events proceed. Eliminated events archived with evidence for later audit.
2 — EM Correlation	Full 10-channel EM record +/- 5 minutes surrounding trigger timestamp	Does optical timestamp match EM threshold crossing within 500ms? Which channels were elevated? What was trigger score composition? What was GPDM state variable profile?	Correlated EM profile: trigger score, channel contributions, 30-day pre-event state variable trajectory, GPDM event class assignment (A through D). A 500ms EM-optical correlation within this context is the core publishable evidence.
3 — Multi-spectrum Fusion	Thermal IR + Visible/NIR + Near-UV + AS7341 — all GPS-timestamped to same trigger	Does IR show thermal anomaly? Does UV show N2/O2 plasma emission lines? What is AS7341 mineral classification? Are IR and visible spatially coincident?	Fused multi-spectrum event profile: thermal anomaly magnitude, visible morphology, UV emission line ID, AS7341 mineral composition. Null results documented per channel — null results are scientifically valuable for eliminating prosaic sources.
4 — GPDM Pathway Classification	Fused EM + optical profile + site geological data + external catalog cross-reference	Assign probable GPDM pathway based on active channels. Cross-reference with USGS seismic catalog and NASA DONKI CME catalog for external forcing context.	Pathway classification with confidence level, external forcing context (seismic activity within 72 hours, Kp index, recent CME), and recommended additional data collection to strengthen or eliminate pathway assignment.
5 — Dashboard Upload	Complete classified event package	Compress and upload full correlated dataset. Generate automated summary for human review queue. Flag high-confidence events for priority review.	Complete event package in human review queue: EM record, fused optical imagery, pathway classification, external forcing context, elimination log, confidence score. Human reviewer determines publication readiness.

9. LIMITATIONS

The Field Node v2.0 extends the EM Sentinel platform and inherits all limitations described in Carter (2026a). Additional constraints apply to the triggered acquisition architecture. The 60-point trigger threshold was derived from weighted theoretical pathway indicators and has not been validated against a confirmed APP event dataset; the threshold may require site-specific tuning, and both false positive activations and missed sub-threshold events should be anticipated during initial deployment. The motor mount targets a slew time under 3 seconds but directional estimation from dual stake pairs carries +/-30-45 degrees of uncertainty sufficient to miss a compact or fast-moving event in the directed camera field of view; the wide-field context camera partially mitigates this. The AI classification pipeline is a proposed architecture requiring training data from confirmed events that does not yet exist, meaning early deployments rely on human review for all classification decisions. The complete system at \$1,788 USD is a significant cost increase over the EM Sentinel alone, and the Sionyx Aurora Pro (\$599) is the largest single cost driver. Power draw during acquisition (~25W peak) requires the upgraded 100W solar panel and 40Ah battery; undersized power systems will cause acquisition failures during extended cloudy periods.

10. CONCLUSIONS

The Presignal Field Node v2.0 represents, to our knowledge, the first integrated triggered multi-spectrum acquisition system designed specifically for geological plasma discharge research. By making thermal IR a primary detection channel rather than a supplementary one — directly addressing the Galan Santos and Koren (2024) finding that most discharge events are invisible at visible wavelengths — the system is designed to capture the events that existing camera deployments most commonly miss.

The data yield framework in Section 4.1 defines what scientific output the instrument produces under each event scenario, from sub-visual EM-only events through full four-domain visual confirmation. Every trigger activation

produces a defined and useful dataset. Background deployment periods produce a scientific baseline independent of event capture. The AI pipeline in Section 8 ensures every capture event produces a fully correlated, pathway-classified package ready for human review.

The open-access release under CC BY 4.0 is intended to enable replication by any research group, citizen science network, or individual investigator. The dual-node Hessdalen configuration in Section 7 is the primary proposed deployment for first publication of triangulated APP source localization data. Researchers interested in collaboration or data contribution to the Presignal monitoring network are invited to contact the author.

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